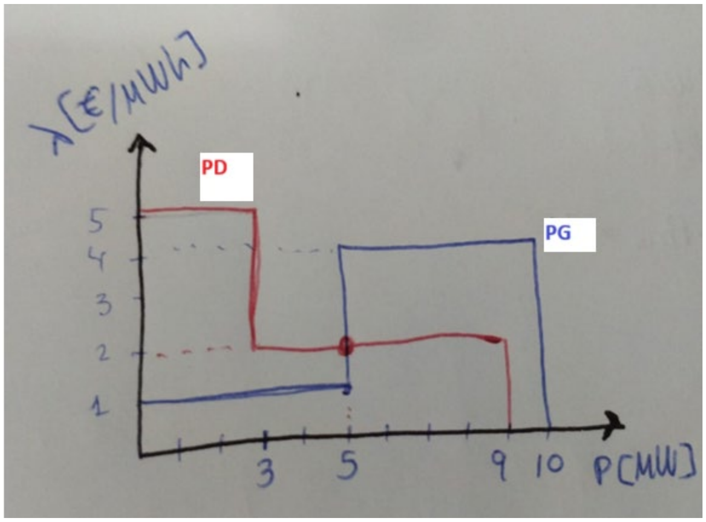
Exercise 5: (SPA) model 2.

Let's consider the instance of the single-producer single-consumer single-period auction model (SPA^0) associated to the following data:

- Generation units: $\mathcal{G} = \{1\},\ \mathcal{B}_G^b = \{1,2\},\ P_G^b = [5 \quad 5]',\ \lambda_G^b = [1 \quad 3]'$
- Demand units: $\mathcal{D} = \{1\},\ \mathcal{B}_D^b = \{1,2\},\ P_D^b = [3 \quad 6]',\ \lambda_D^b = [5 \quad 2]'$,

a) Draw the aggregated generation and demand bid functions and find graphically the market equilibrium point



Equilibrium point at Price= 2€/MWh and Power=5MW

P11**D**= 3 MW (lleno)
P21**D**= 2 MW (medio lleno)
P11**G**= 5 MW (lleno)
P21**G**= 0 MW (vacío)

b) Write down the explicit mathematical formulation of the associated single-period auction optimization model (SPA^0) that corresponds to the data of this instance.

$$\max_{P_G, P_D} SW(P_G, P_D) = 5 \cdot P_{11D} + 2P_{21D} - P_{11G} - 3P_{21G}$$

subject to:

$$P_{11D} \leq 3 \quad \text{Matched\_D}$$
$$P_{21D} \leq 6$$
$$P_{11G} \leq 5 \quad \text{Matched\_G}$$
$$P_{21G} \leq 5$$

$$P_{11D} + P_{21D} - P_{11G} - P_{21G} = 0 \quad \text{Market}$$

$$P_{11D}, P_{21D}, P_{11G}, P_{21G} \geq 0$$

c) Write down the necessary and sufficient optimality conditions (KKT) for this (SPA^0) instance.

$$
kkT1 \begin{cases} \lambda^* = \lambda^b_{G1k} - \mu^0_{G1k} - \mu^b_{G1k} \\ \lambda^* = \lambda^b_{D1k} + \mu^0_{D1k} + \mu^b_{D1k} \end{cases}
\qquad
kkT2 \begin{cases} \mu_i g_i(x^*) = 0 \end{cases}
$$

$$
kkT1 \begin{cases} 1 - \mu^0_{G11} - \mu^b_{G11} - \lambda^* = 0 \\ 3 - \mu^0_{G12} - \mu^b_{G12} - \lambda^* = 0 \\ 5 + \mu^0_{D11} + \mu^b_{D11} - \lambda^* = 0 \\ 2 + \mu^0_{D12} + \mu^b_{D12} - \lambda^* = 0 \end{cases}
$$

P_G^0

$$
kkT2 \begin{cases} \mu^b_{G11}(P_{G11} - 5) = 0 \\ \mu^b_{G12}(P_{G12} - 5) = 0 \\ \mu^b_{D11}(P_{D11} - 3) = 0 \\ \mu^b_{D12}(P_{D12} - 6) = 0 \\ \mu^0_{G11} P_{G11} = 0 \\ \mu^0_{G12} P_{G12} = 0 \\ \mu^0_{D11} P_{D11} = 0 \\ \mu^0_{D12} P_{D12} = 0 \end{cases}
$$

$$
kkT3 \longrightarrow \mu^b_G, \mu^b_D \leq 0 \quad . \ \mu^0_G \mu^0_D \geq 0.
$$

d) With the help of the (KKT) conditions prove that the equilibrium point found in a) is the global optimal solution of this (SPA^0) instance.

equilibrium point. $P_{G11}^* = 5$ $P_{D11}^* = 3$ $\quad X^* = 2.$

$P_{G12}^* = 0$ $P_{D12}^* = 2$

Using KKT2

$\mu_{G11}^b (P_{G11} - 5) = 0$ $\quad \mu_{G11}^b = ?$

$\mu_{G12}^b (P_{G12} - 5) = 0$ $\quad \boxed{\mu_{G12}^b = 0}$

$\mu_{D11}^b (P_{D11} - 3) = 0$ $\quad \mu_{D11}^b = ?$

$\mu_{D12}^b (P_{D12} - 6) = 0$ $\quad \boxed{\mu_{D12}^b = \emptyset}$

$\mu_{G11}^0 P_{G1} = \emptyset$ $\quad \boxed{\mu_{G11}^0 = \emptyset}$

$\mu_{G12}^0 \cdot P_{G12} = \dots$

$\mu_{D11}^0 \cdot P_{D11} = \emptyset$ $\quad \boxed{\mu_{D11}^0 = \emptyset}$

$\mu_{D12}^0 \cdot P_{D12} = \emptyset$ $\quad \boxed{\mu_{D12}^0 = \emptyset}$

Using KKT1

$1 - \cancel{\mu_{G11}} - \mu_{G11}^b - 2 = 0$ $\quad \boxed{\mu_{G11}^b = -1}$

$3 - \mu_{G12}^0 - 2 = 0$ $\quad \boxed{\mu_{G12}^0 = 1}$

$5 + \mu_{D11}^b - 2 = 0$ $\quad \boxed{\mu_{D11}^b = -3}$

$2 + -2 = \emptyset$ ✓

✓ KKT3

$\begin{pmatrix} \mu_{G11}^0 & \mu_{G12}^0 \\ \mu_{D11}^0 & \mu_{D12}^0 \end{pmatrix} \geq 0$ ✓ $\quad \begin{pmatrix} \emptyset & 1 \\ \emptyset & \emptyset \end{pmatrix}$

$\begin{pmatrix} \mu_{G11}^b, & \mu_{G12}^b \\ \mu_{D12}^b & \mu_{D11}^b \end{pmatrix} \leq 0$ ✓ $\quad \begin{pmatrix} -1 & \emptyset \\ -3. & \emptyset \end{pmatrix}$

solution of problem

$\boxed{X^* = 2 \quad P^* = 5MW}$

e) Find the optimal market clearing with GAMS. Identify the value of the optimal matched bids P_G^* and P_D^* and Lagrange multipliers $\lambda^*, \mu_G^{0^*}$, $\mu_G^{b^*}, \mu_D^{0^*}$ and $\mu_D^{b^*}$and check that they coincides with the values found in sections a) and d).

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----      165 PARAMETER total_matched_demand

              SPA0

D1           5.000


----      165 PARAMETER market_equilibrium

                  SPA0

SocWel          14.000
price            2.000
energy           5.000


----      165 PARAMETER muG_0

               la^*=        la^b        -mu^0        -mu^b

G1.1           2.000        1.000                    1.000
G1.2           2.000        3.000       -1.000
                                                     muG11
                                        muG12
----      165 PARAMETER muD_0

               la^*=        la^b        +mu^b

D1.1           2.000        5.000       -3.000      muD11
D1.2           2.000        2.000
```

f) Check that the optimal solution found by GAMS satisfies the Market Equilibrium Conditions (7) and (8).

First market equilibrium condition ok,(3+2)-5=0

$$\sum_{i\in\mathcal{G}}\sum_{k\in\mathcal{B}_i^b} P_{G_{ik}} - \sum_{i\in\mathcal{D}}\sum_{k\in\mathcal{B}_i^b} P_{D_{ik}} = 0 \qquad \textbf{MARKET}$$

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  SW  Social welfare (objective function)

---- VAR PG  Matched generation of each block [MW]

            LOWER          LEVEL          UPPER         MARGINAL

G1.1          .            5.0000         +INF             .
G1.2          .              .            +INF          -1.0000

---- VAR PD  Matched demand of each block [MW]

            LOWER          LEVEL          UPPER         MARGINAL

D1.1          .            3.0000         +INF             .
D1.2          .            2.0000         +INF             .
```

Second condition also check

$$P_G, P_D \geq 0$$